THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 5 is: 1

(a) $\frac{15}{20}$ **(b)** $\frac{242}{1024}$ **(c)** $\frac{243}{1024}$ **(d)** $\frac{1}{1024}$

2. A cube of side 6 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 96 **(b)** 8 **(c)** 48 **(d)** 64

3. The net of a square pyramid consists of: 1

(a) one square and four rectangles **(b)** two squares and four triangles
(c) one square and four triangles **(d)** five squares

4. The top view of a cube is a: 1

(a) triangle **(b)** hexagon **(c)** circle **(d)** square

5. A polyhedron has 24 vertices and 36 edges. The number of faces is: 1

(a) 58 **(b)** 14 **(c)** 62 **(d)** 12

6. A Koch curve starts with one segment. The number of segments at stage 5 is: 1

(a) 243 **(b)** 1023 **(c)** 1024 **(d)** 20

7. A cube of side 7 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 150 **(b)** 125 **(c)** 8 **(d)** 60

8. The net of a regular tetrahedron consists of: 1

(a) two triangles and two squares **(b)** four equilateral triangles
(c) three equilateral triangles **(d)** four squares

9. The cross-section of a cube cut by a plane parallel to one of its faces is a: 1

(a) triangle **(b)** hexagon **(c)** square **(d)** circle

10. A polyhedron has 60 vertices and 90 edges. The number of faces is: 1

(a) 32 **(b)** 148 **(c)** 152 **(d)** 30

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid has 12 faces and 30 edges. Use Euler's formula to find its number of vertices. 2

12. A prism has 42 edges. Find the number of sides of its base and the number of faces. 2

13. Verify Euler's formula for a 14-sided pyramid. 2

14. A solid has 14 faces and 36 edges. Use Euler's formula to find its number of vertices. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 7 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3
(i) exactly two faces painted
(ii) exactly one face painted.

16. A Koch curve starts with a straight line of length 54 cm. How many segments are there at stage 3, how long is each, and what is the total length? 3

17. A Sierpinski triangle starts with a triangle of area 1024 cm². Find the shaded area after stage 3 and the area removed. 3

OR

A cube of side 8 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3
(i) exactly two faces painted
(ii) exactly one face painted.

18. A Koch curve starts with a straight line of length 162 cm. How many segments are there at stage 4, how long is each, and what is the total length? 3

OR

A Sierpinski triangle starts with a triangle of area 1024 cm². Find the shaded area after stage 4 and the area removed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the number of faces, vertices and edges of 5
(i) a heptagonal prism and
(ii) a heptagonal pyramid, and verify Euler's formula for both.

OR

A cube of side 20 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 8000. 5

20. Find the number of faces, vertices and edges of 5
(i) a octagonal prism and
(ii) a octagonal pyramid, and verify Euler's formula for both.

OR

A cube of side 25 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 15625. 5

END OF QUESTION PAPER — Total: 40 Marks